PREDICTING DIGITAL EYE STRAIN & VISION RISK USING LIFESTYLE BEHAVIORAL INDICATORS

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ABSTRACT

In the post-pandemic era, digital device usage has become common across all people, from teenagers to working professionals and seniors. This increased screen time of the devices has led to a surge in Computer Vision Syndrome (CVS). While existing research often focuses on specific groups (e.g. office workers or students) or uses complex medical imaging, this project proposes a widely accessible, preventive approach. This study aims to predict the risk of developing digital eye strain based on lifestyle and behavioral habits rather than just clinical diagnostics. Using a self-conducted survey via Google Forms, data was collected on factors such as screen time, viewing distance, lighting conditions, hydration, and blink awareness. Machine learning models, specifically Random Forest (RF) and Decision Tree (DT), were trained to classify individuals as "At Risk" or "No Risk." The Random Forest model achieved an accuracy of 61.90%, outperforming the Decision Tree (57.14%). The study highlights that non-visual factor, such as water intake and conscious blinking, play a crucial role in predicting eye strain, offering a basis for personalized preventative health tools.

Keywords: Machine Learning, Random Forest, Decision Tree, Supervised Learning, Computer Vision Syndrome (CVS).

I. INTRODUCTION

1.1 Background of the Study

The 21st century has witnessed a paradigm shift in how humans interact with information. From remote work to digital education, screen dependency has become common across all age groups. However, this physiological shift has outpaced our biological adaptation, leading to a global rise in Computer Vision Syndrome (CVS), also known as Digital Eye Strain. Unlike traditional optical issues caused by aging or genetics, CVS is a lifestyle disorder characterized by dry eyes, blurred vision and headaches.

1.2 Problem Statement

Current diagnostic methods for CVS are reactive rather than preventative. Patients typically seek help only after symptoms become severe. Furthermore, most existing predictive models are trained on specific, homogenous datasets (e.g., IT professionals or medical students), making them inaccurate for the general population. There is a critical lack of tools that analyze "micro-habits" such as blink awareness and hydration alongside standard metrics like screen time.

1.3 Research Objectives

The primary objectives of this study are:

1. To construct a diverse dataset capturing digital habits across a general age group (18–60+ years).
2. To identify the correlation between non-visual physiological factors (e.g., hydration, water intake) and eye strain.
3. To develop and evaluate Random Forest Tree and comparing its performance against models like Logistic Regression and Decision Trees.

1.4 Significance of the Study

This research bridges the gap between optometry and data science. By identifying modifiable risk factors, the proposed model empowers users to make lifestyle changes before clinical intervention is required. It shifts the focus from "Diagnosis" to "Prognosis and Prevention."

II. LITERATURE REVIEW

2.1 Existing Approaches

Existing research on Computer Vision Syndrome largely focuses on occupational health (office workers) or uses image processing techniques. Below is a summary of relevant studies, highlighting the gaps this research attempts to fill.

Doi	Name of the Paper	Year	Algorithm Used	Accuracy	Future Scope	Research Gap
10.1177/20552076251382845	Exploratory ML models for CVS in occupational health	2025	XGBoost, RF, SVM	70.6%	Larger cohorts needed to confirm scalability	Focuses heavily on demographics (age/sex) rather than modifiable lifestyle habits.
10.1109/ACCESS.2023	Real-time Eye Blink Detection for Strain Prediction	2023	CNN (Image Processing)	92%	Integration into mobile apps for real-time alerts.	Requires active camera monitoring; does not account for hydration or lighting.
10.1007/s12652-021-035	Predictive Analysis of CVS among	2022	Naïve Bayes, J48	85.4%	Inclusion of psycholog	Limited to software professionals;

	Software Professionals				ical stress factors.	ignores recreational mobile usage.
10.1016/j.procs.2021	Machine Learning Approach for Detecting Eye Fatigue	2021	SVM, KNN	88%	Wearable sensors for better data collection.	Relies on physiological sensors, not accessible self-reported data.
10.3390/ijerph18168663	Association of Screen Time and Dry Eye Disease	2021	Logistic Regression	N/A (Statistical)	Long-term longitudinal studies.	Statistical correlation only; no predictive ML model created.
10.1007/978-981-15	Smart Healthcare Monitoring for Eye Strain	2020	Artificial Neural Networks	91%	IoT based smart glasses integration.	High cost of implementation; requires hardware.
10.1051/e3sconf/202343001036	Predicting Smartphone Vision Syndrome: A Feasible Approach using Machine Learning Algorithms	2023	RF, SVM, Naïve Bayes	97%	Expanding the dataset, integrating real-time behavioral monitoring, and deploying the model.	No real-time monitoring or preventive applications

2.2 Unique Contribution of This Study

Most existing research relies on image processing (detecting blinks via camera) or occupational surveys (focusing on office).

This study is unique because:

1. **Behavioral Focus:** It incorporates overlooked lifestyle variables such as Water Intake (hydration affects tear production) and Conscious Blink Awareness.

2. **Preventative Approach:** Unlike camera-based systems that detect strain after it happens, this model predicts risk based on habits, allowing users to change their behavior before symptoms worsen.

3. **Accessibility:** It requires no special hardware (cameras/sensors), making the prediction accessible to anyone with a smartphone.

III. THEORETICAL FRAMEWORK

The study is grounded in the physiological theory of eye strain.

- **The Tear Film Theory:** Adequate hydration and frequent blinking are required to maintain the mucin, aqueous, and lipid layers of the eye. Screen use reduces blink rate by up to 66%, leading to dry eyes.
- **The Accommodation Theory:** The ciliary muscles in the eye must contract to focus on near objects (screens). Prolonged contraction without breaks leads to fatigue.
- **The Blue Light Hypothesis:** High-energy visible (HEV) light from screens may contribute to retinal stress and sleep disruption, strain symptoms.

By quantifying these factors (Distance, Lighting, Water Intake, Screen Time) as input features, we can theoretically predict the physiological outcome (Symptoms).

IV. METHODOLOGY

4.1 Data Acquisition

Data was collected using a structured Google Form survey distributed to a diverse group of participants (students and working professionals). The survey consisted of four sections:

1. **Basic Information:** Age, Gender.
2. **Digital Device Habits:** Primary device, screen time, continuous usage, distance, lighting, and use of "Night Mode."
3. **Biological Factors:** Parental history of glasses, daily water intake, and blink awareness.
4. **Current Eye Health:** Current glasses usage and frequency of symptoms (Headache, Dry Eyes, Blurred Vision).

4.2 Data Dictionary (Input Features)

The dataset consists of 14 features mapped to a binary target.

Variable	Type	Description
Age	Numeric	User's age (Standard Scaled)
Screen_Time	Numeric	Average daily hours (Extracted via Regex)
Continuous_Time	Ordinal	Longest single session without breaks
Distance	Categorical	Viewing distance (<30cm, Arm's Length, >50cm)
Lighting	Categorical	Dark, Dim, or Well-lit room
Hydration	Ordinal	Daily water intake (<1L to >2L)

Blink_Aware	Binary	Conscious effort to blink (Yes/No)
Genetics	Ordinal	Parental history of glasses
Symptoms	Target	Derived from frequency of symptoms (0 vs 1)

4.3 Data Preprocessing (Python Implementation)

The raw data was cleaned and transformed using Python (Pandas).

- **Feature Selection:** Irrelevant columns like Timestamp and Email were dropped.
- **Label Encoding:** Categorical text data was converted into numerical format for machine learning.
  - *Example:* 'Male' → 0, 'Female' → 1.
  - *Example:* 'Well-lit room' → 0, 'Dim light' → 1, 'Dark room' → 2.
- **Target Variable:** The Symptoms column was binary encoded where 'Never/Rarely' = 0 (No Risk) and 'Sometimes/Frequently' = 1 (Risk).
- **Splitting:** The dataset was split into **80% training** and **20% testing** sets to evaluate performance on unseen data.

4.4 Model Implementation

Two supervised machine learning algorithms were implemented using Scikit-Learn:

1. **Random Forest Classifier (RF):** An ensemble learning method that constructs multiple decision trees during training.
   - *Parameters:* n_estimators=200, max_depth=4.
2. **Decision Tree Classifier (DT):** A tree-structured classifier that splits data based on feature values.
   - *Parameters:* criterion='gini', max_depth=3.

V. RESULTS AND ANALYSIS

The models were evaluated based on Accuracy, Precision, Recall, and F1-Score.

5.1 Random Forest Results

- Accuracy: 61.90%
- Precision: 62.50%
- Recall: 50.00%
- F1 Score: 55.56%

Analysis: The Random Forest model showed a balanced performance. It was slightly better at precision, meaning when it predicted a user was "At Risk," it was often correct. However, the recall was 50%, indicating it missed half of the actual risk cases.

5.2 Decision Tree Results

- Accuracy: 57.14%
- Precision: 57.14%
- Recall: 72.73%
- F1 Score: 64.00%

Analysis: The Decision Tree had lower overall accuracy but a significantly higher Recall (72.73%). In the context of health predictions, high recall is often preferred because it is better to falsely warn a healthy user (False Positive) than to miss a user who is actually at risk (False Negative).

5.3 Visualization

A scatter plot comparing Screen Time vs. Continuous Usage Time revealed a clear trend: users with high continuous usage (>2 hours) without breaks were mainly in the "Risk" category (Class 1), validating the "20-20-20 rule" (take a break every 20 minutes).

VI. DISCUSSION

6.1 Why Random Forest Performed Better in Accuracy

Random Forest generally outperforms single Decision Trees because it aggregates the results of many trees (200 in this study). This "ensemble" approach reduces overfitting and handles noise in the survey data better. The single Decision Tree, while having better recall, was likely less stable and more prone to specific biases in the training data.

6.2 The Role of Lifestyle Features

The study found that features like Water (hydration) and Blink_Aware were significant. This suggests that eye strain is not just about how long you look at a screen, but how you look at it and how you treat your body while doing so. Users who drank less than 1 liter of water and paid no attention to blinking reported higher symptom frequencies.

6.3 Limitations

The primary limitation was the small dataset size (test support = 21 implies a total dataset of ~105 rows). Machine learning models generally require larger volumes of data to generalize well. The accuracy of ~62% is a promising baseline but indicates that more data is needed for a robust clinical tool.

VII. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This research successfully demonstrated that lifestyle behaviors—specifically hydration, lighting, and blink awareness—can be used to predict the risk of Digital Eye Strain using machine learning. While the Random Forest model provided the most stable accuracy (61.90%), the Decision Tree highlighted the importance of catching potential risk cases (High Recall).

7.2 Future Scope

1. **Larger Dataset:** Expanding the survey to 1000+ participants to improve model accuracy.
2. **Real-Time Application:** Developing a desktop or mobile app that runs this model in the background, prompting users to drink water or blink if their risk score gets too high.
3. **Computer Vision Integration:** Combining this lifestyle analysis with a webcam-based blink detector for a hybrid, high-accuracy health monitoring system.

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